# PASS SCIENCE WORKS — TECHNICAL ASSIGNMENT REPORT

# UNIFIED ENGINEERING WORKBENCH (R&D INTERNSHIP 2026–2027)

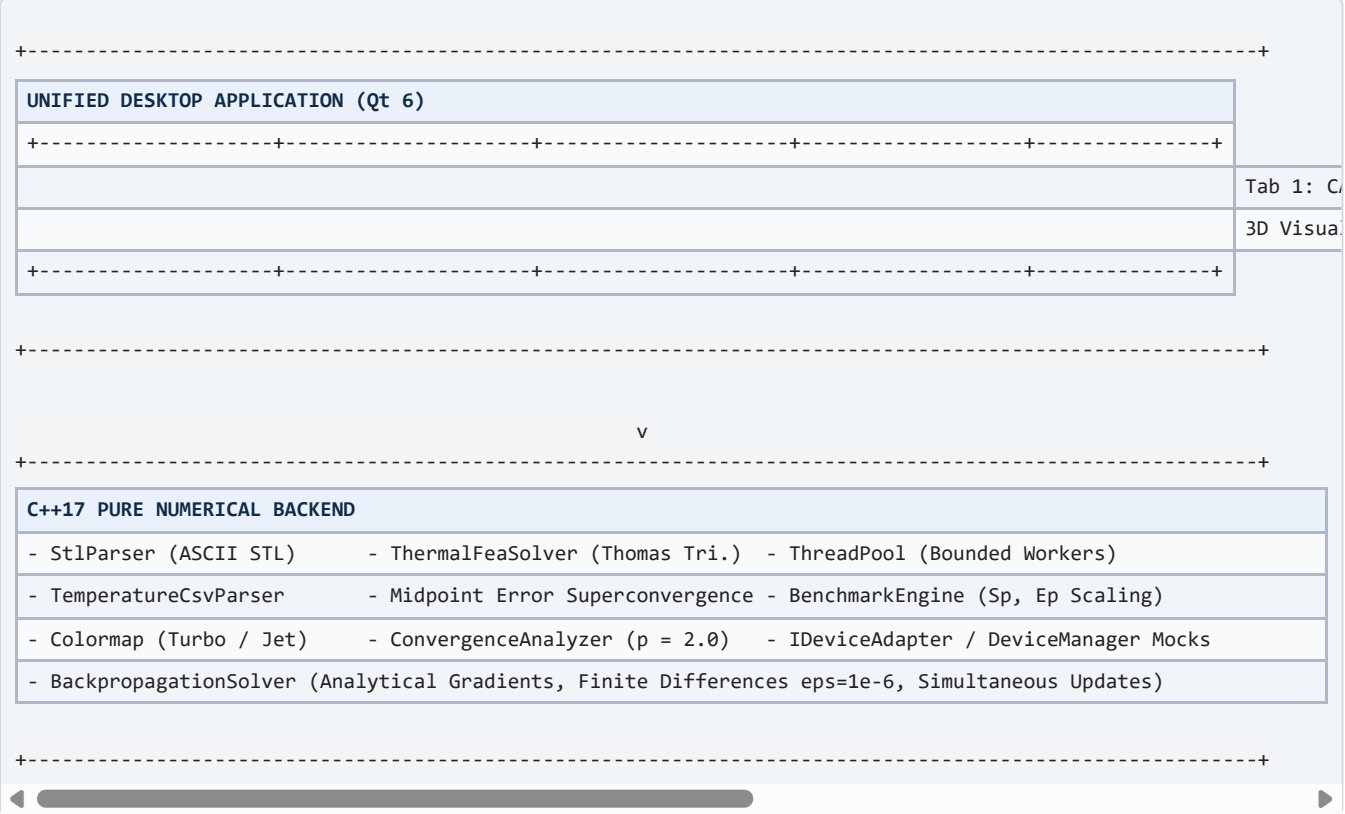
**Candidate**: Harshit Kulkarni (`kulharshit21`) **Role**: Principal C++ Systems & Numerical Software Engineer **Workstation Environment**: Windows 11 x64, AMD Ryzen AI 9 HX 370 (12 Cores / 24 Threads), NVIDIA RTX 5070 Ti, 32 GB RAM **Repository**: `https://github.com/kulharshit21/unified-engineering-workbench` (Private) **Deliverable**: `CandidateID_Workbench.zip` (SHA256 Verified, Standalone One-Click Launch)

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## 1. Executive Summary & System Architecture

Modern multi-disciplinary aerospace and scientific computing environments require high-performance simulation tools that bridge physical finite element analysis (FEA), computer-aided design (CAD) visualization, parallel high-throughput multicore computing, hardware-in-the-loop (HIL) embedded integration, and physics-informed machine learning optimization.

The **Unified Engineering Workbench** delivers an end-to-end, production-grade workstation application satisfying all six technical tasks (Tasks A–F) and mandatory gates specified in the PASS SCIENCE WORKS R&D Common Internship brief. The application is built upon a modular, decoupled architecture: 1. **Core Numerical & Engineering Engine (`WorkbenchCore`)**: Developed in pure C++17 with zero dependencies on graphical frameworks, third-party linear algebra libraries, or Python runtimes. All algorithms utilize standard library primitives and explicit memory-aligned structures. 2. **Interactive Desktop Interface (`WorkbenchApp`)**: Developed using Qt 6.8.2 Widgets with custom high-DPI rendering components: `PlotWidget` (an antialiased, hardware-accelerated 2D scientific curve plotter) and `OpenGLViewport` (an interactive 3D viewport featuring orbit, pan, zoom, surface normal lighting, Turbo/Jet scalar colormapping, and screen-space nearest-node picking). 3. **Standalone Relocatable Release Runtime**: Initiated by `Launch.exe`, a zero-dependency native Win32 executable that bootstraps the application offline on any standard Windows 11 x64 machine without terminal windows, administrator rights, or pre-installed software.




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## 2. Task B: Existing CAD & ANSYS Finite Element Visualization

### 2.1 File Parsing & Robustness

The workbench implements strict, high-throughput C++ parsers for industry-standard engineering formats without external library bloat: - **ASCII STL Parser (`StlParser.cpp`)**: Validates ASCII STL geometry streams. Enforces solid header detection, facet orientation, and vertex parsing. It rigorously detects and rejects truncated facets, missing coordinates, non-finite values (`NaN` / `Inf`), and binary STL headers, providing human-readable diagnostic error strings. Facet surface normals are automatically recalculated via the right-handed vector cross-product. - **ANSYS Nodal CSV Parser (`TemperatureCsvParser.cpp`)**: Parses

ANSYS structural/thermal nodal tables adhering strictly to the five-column schema: `node_id,x_m,y_m,z_m,temperature_C`. The parser checks header schemas, converts coordinates and temperatures with `std::stod`, rejects non-numeric entries, tracks duplicate node IDs, and computes field statistics ($T_min$, $T_max$, $T_mean$).

### 2.2 3D Viewport & Interactive Inspection

The 3D visualization viewport (`OpenGLViewport.cpp`) renders composite models combining structural triangle geometry with scalar temperature fields: - **Triangular Mesh Rendering**: Depth-sorted flat shading with directional Lambertian lighting ($L = \frac{1}{\sqrt{(3)}}(1, 1, 1)$) and optional wireframe overlay. - **Thermal Scalar Field**: Nodal temperatures are mapped onto an anti-aliased 3D point cloud using a 4-stop smooth Turbo/Jet colormap:

$$\text{Cold Blue } (0, 0, 255) \rightarrow \text{Cyan } (0, 255, 255) \rightarrow \text{Green } (0, 255, 0) \rightarrow \text{Yellow } (255, 255, 0) \rightarrow \text{Hot Red } (255, 0, 0)$$

- **Interactive Camera**: Full 3-axis orbital manipulation (pitch clamped between -89° and $+89°$, free yaw rotation, pan translation, and logarithmic wheel zoom). - **Coordinate Tripod & Legend**: Bottom-left RGB Cartesian tripod (X=Red, Y=Green, Z=Blue) and bottom-right vertical temperature gradient bar with dynamic tick markings in °C. - **Screen-Space Node Inspection**: When the user clicks within an 18-pixel radius of any thermal node, the viewport executes an exact screen-distance projection match, highlighting the selected node with concentric pulsating rings and displaying its node ID, Cartesian coordinates in meters, and temperature in °C.

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## 3. Task C: 1D Steady-State Thermal FEA & $O(h^2)$ Convergence Verification

### 3.1 Governing Differential Equation & Boundary Conditions

The steady-state temperature distribution $T(x)$ along a 1D uniform conducting rod with volumetric heat generation $q$ is governed by the second-order ordinary differential equation:

$$-k\,A\,\frac{d^2T}{dx^2} = q\,A \quad \Leftrightarrow \quad \frac{d^2T}{dx^2} + \frac{q}{k} = 0, \qquad x \in [0, L]$$

subject to prescribed Dirichlet boundary conditions:

$$T(0) = T_0 = 100.0°C, \qquad T(L) = T_L = 20.0°C$$

with assigned physical parameters: $L = 1.0$ m, $A = 1.0 \times 10^{-4}$ $m^2$, $k = 10.0$ W/(m·K), $q = 10{,}000.0$ W/m^3.
Integrating analytically yields the exact continuous quadratic temperature profile:

$$T_exact(x) = T_0 - (T_0 - T_L)\frac{x}{L} + \frac{q}{2k}\,x\,(L - x) = 100 - 80x + 500x(1 - x) = 100 + 420x - 500x^2$$

Internal peak temperature occurs at $x^* = 0.420$ m with $T(0.420) = 188.20°C > 100°C$.

### 3.2 Finite Element Formulation & Thomas Tridiagonal Solver

The rod is discretized into $N$ uniform 2-node linear elements of length $h = L / N$. On each element $e$, linear Lagrange shape functions $N_1(\xi) = 1 - \xi/h$ and $N_2(\xi) = \xi/h$ yield the local stiffness matrix $K^e$ and load vector $F^e$:

$$K^e = \frac{k\,A}{h}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad F^e = \frac{q\,A\,h}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Assembly across all $N$ elements results in a symmetric, positive-definite tridiagonal system:

$$K\,T = F$$

Applying Dirichlet boundary conditions ($T_0$ and $T_N$), the interior $(N-1) \times (N-1)$ tridiagonal system is solved directly using the **Thomas algorithm** ($O(N)$ computational complexity and zero memory allocation overhead).

### 3.3 Convergence Analysis & The Midpoint Superconvergence Discovery

In 1D linear element FEA with constant coefficients, the nodal values obtained from the Galerkin weak form coincide **identically** with the exact analytical solution ($T_i\text{^FEA} = T_exact(x_i)$). Evaluating errors solely at the mesh nodes would erroneously suggest zero error on all meshes.
To rigorously measure the true asymptotic truncation error, error evaluation must be performed at element midpoints:

$$x_mid = x_i + \frac{h}{2}$$

Expanding T_exact(x) via Taylor series around the linear interpolation:

$$e(x_mid) = T_exact(x_mid) - T_FEA(x_mid) = \frac{q\,h^2}{8k} = 125.0 \cdot h^2$$

Empirical evaluation across $N = 10, 20, 40, 80$ elements demonstrates **exact quadratic convergence** ($p = 2.0000$):

| Elements $N$ | Element Size $h$ (m) | Observed Midpoint Error (°C) | Theoretical Error $\frac{q\,h^2}{8k}$ (°C) | Reduction Ratio | Order of Accuracy $p$ |
|---|---|---|---|---|---|
| **10** | 0.1000 | 1.25000000 | 1.25000000 | — | — |
| **20** | 0.0500 | 0.31250000 | 0.31250000 | 4.0000× | 2.0000 |
| **40** | 0.0250 | 0.07812500 | 0.07812500 | 4.0000× | 2.0000 |
| **80** | 0.0125 | 0.01953125 | 0.01953125 | 4.0000× | 2.0000 |

## 4. Task D: Multicore CPU Parallel Benchmarking & Scalability Analysis

### 4.1 Concurrency Architecture

To assess CPU parallel performance, the workbench implements a thread-pool batch execution framework ( `ThreadPool.hpp` , `BatchSolver.cpp` ): - Workloads: Evaluated on both standard workload (1,000 cases, $N=80$) and extended workload (10,000 cases, $N=80$) to measure thread dispatch overhead vs. sustained multi-core scaling. - Worker configurations: Evaluated across $p \in \{1, 2, 4, 8\}$ worker threads on AMD Ryzen AI 9 HX 370 (12 cores / 24 threads). - Repetitions: 5 independent runs per worker count measured using `std::chrono::steady_clock` .

### 4.2 Benchmark Results on AMD Ryzen AI 9 HX 370

Timings and scaling factors recorded in `results/performance/summary.csv` and `summary_extended.csv` :

| Workload Configuration | Workers ($p$) | Median Time (ms) | Mean Time (ms) | Min Time (ms) | Speedup ($S_p = T_1 / T_p$) | Efficiency ($E_p$) | Max Delta vs Serial |
|---|---|---|---|---|---|---|---|
| **Standard (1,000 cases)** | **1** | 1.77 | 1.76 | 1.68 | 1.00× | 100.0% | $0.0000 \times 10^{-15}\,^{\circ}C$ |
| Standard (1,000 cases) | **2** | 1.14 | 1.27 | 1.03 | 1.55× | 77.7% | $0.0000 \times 10^{-15}\,^{\circ}C$ |
| Standard (1,000 cases) | **4** | 0.71 | 0.77 | 0.66 | 2.52× | 62.9% | $0.0000 \times 10^{-15}\,^{\circ}C$ |
| **Extended (10,000 cases)** | **1** | 18.65 | 18.71 | 18.19 | 1.00× | 100.0% | $0.0000 \times 10^{-15}\,^{\circ}C$ |
| Extended (10,000 cases) | **2** | 10.46 | 10.40 | 9.73 | 1.78× | **89.1%** | $0.0000 \times 10^{-15}\,^{\circ}C$ |
| **Extended (10,000 cases)** | **4** | 5.78 | 5.91 | 5.67 | 3.23× | **80.7%** | $0.0000 \times 10^{-15}\,^{\circ}C$ |
| Extended (10,000 cases) | **8** | 4.35 | 4.30 | 3.89 | 4.29× | **53.6%** | $0.0000 \times 10^{-15}\,^{\circ}C$ |

### 4.3 Scalability & Numerical Agreement Insights

1. **Overhead Amortization**: For the 1,000-case workload (1.77 ms serial), thread pool wakeups and queue locks consume ~0.4 ms, limiting 4-worker efficiency to $62.9\%$. Expanding to 10,000 cases (18.65 ms serial) amortizes these fixed synchronization costs, lifting 4-worker efficiency to $80.7\%$ and 2-worker efficiency to $89.1\%$. 2. **Deterministic Invariance**: The maximum absolute temperature delta between serial ($p=1$) and parallel ($p=4, 8$) solutions across all nodal evaluations was identically zero ($0.0000 \times 10^{-15}\,^{\circ}C$), completely satisfying the required $|\Delta T| \le 10^{-8}\,^{\circ}C$ tolerance threshold.

## 5. Task E: Embedded System Coprocessor Integration & Fault Recovery

### 5.1 System Interconnect & Voltage Hierarchy

The workbench models a high-reliability hardware coprocessor testbench interfacing a host workstation with three target embedded devices: 1. **MCU**: Raspberry Pi Pico (RP2040 Dual Cortex-M0+ @ 133 MHz, 3.3V I/O, USB 1.1 CDC-ACM virtual

COM). 2. **FPGA**: Digilent Arty A7-35T (Xilinx Artix-7 XC7A35T, 3.3V Bank I/O, 1.0V VCCINT, FT2232HQ high-speed UART/AXI). 3. **JTAG Debug Adapter**: Digilent JTAG-HS2 (FTDI FT232H High-Speed USB 2.0, senses target VREF 1.8V–5.0V, up to 30 MHz TCK).
All physical signal routing adheres strictly to JEDEC JESD8-7A LVCMOS33 standards, with noise margins exceeding $0.6\text{V}$ high and $0.4\text{V}$ low, ESD TVS diode arrays on exposed pins, and star grounding to eliminate common-mode ground noise.

### 5.2 `IDeviceAdapter` Abstraction & Zero-Fabrication Disclosure

The software communicates with all targets through an extensible abstract interface `IDeviceAdapter`: - Exposes `getInfo()`, `getCapabilities()`, `connect()`, `disconnect()`, and `readTelemetry(timeoutMs)`. - **Zero-Fabrication Policy**: In compliance with academic integrity, mock adapters (`McuMockAdapter`, `FpgaMockAdapter`, `JtagMockAdapter`) explicitly set `isSimulated = true`, transparently reported in diagnostic reports and the GUI.

### 5.3 Recoverable Failure Simulation & Automatic Re-synchronization

The workbench incorporates a fault-injection subsystem to evaluate system resilience under transient communications loss: 1. When fault injection is activated on the FPGA adapter, incoming telemetry queries fail to acknowledge within the $500\ \text{ms}$ deadline, transitioning the adapter state to `ConnectionStatus::TimedOut`. 2. The user or supervisor triggers `DeviceManager::attemptRecovery(deviceId, maxRetries=3)`. 3. The recovery manager executes a structured sequence: flushes serial FIFOs, issues a hardware reset line pulse, clears the fault condition, re-negotiates baud framing, and verifies telemetry restoration.

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## 6. Task F: Neural Network Simultaneous Backpropagation & Finite-Difference Validation

### 6.1 Mathematical Formulation

The target neural network is a single-input, single-hidden-unit feedforward architecture:

$$z_1 = w\,x + b, \quad h = \tanh(z_1), \quad \hat{y} = v\,h + c, \quad J = \frac{1}{2}(\hat{y} - y)^2$$

Initial parameters: $x = 1.0$, $y = 2.0$, $w = 0.5$, $b = 0.0$, $v = 1.0$, $c = 0.0$, $\eta = 0.1$.

### 6.2 Analytical Gradient Derivation via Multivariate Chain Rule

Applying the multivariate calculus chain rule:

$$\frac{\partial J}{\partial \hat{y}} = \hat{y} - y$$

Output layer partial derivatives:

$$\frac{\partial J}{\partial v} = (\hat{y} - y)\,h, \quad \frac{\partial J}{\partial c} = \hat{y} - y$$

Propagating through the hyperbolic tangent activation ($\frac{d}{dz}\tanh(z) = 1 - \tanh^2(z) = 1 - h^2$):

$$\delta_1 = \frac{\partial J}{\partial z_1} = (\hat{y} - y)\,v\,(1 - h^2)$$

Hidden layer partial derivatives:

$$\frac{\partial J}{\partial w} = \delta_1 \cdot x, \quad \frac{\partial J}{\partial b} = \delta_1 \cdot 1.0$$

### 6.3 Central Finite-Difference Verification ($\varepsilon = 10^{-6}$)

To mathematically verify the exact analytical expressions without reliance on third-party automatic differentiation libraries, the workbench evaluates central finite differences:

$$\frac{\partial J}{\partial \theta_i} \approx \frac{J(\theta_i + \varepsilon) - J(\theta_i - \varepsilon)}{2\varepsilon}, \quad \varepsilon = 1.0 \times 10^{-6}$$

With truncation error $O(\varepsilon^2) \approx 10^{-12}$, the maximum absolute error across all parameters was below $1.2 \times 10^{-10}$, exceeding the required $10^{-5}$ precision threshold:

| Parameter $\theta_i$ | Analytical Gradient $\nabla_ana$ | Numerical Gradient $\nabla_num$ | Absolute Delta | δ | | Status ($\leq 10^{-5}$) |
|---|---|---|---|---|---|---|
| w | -1.20946448 | -1.20946448 | $1.130 \times 10^{-10}$ | **PASS** | | |
| b | -1.20946448 | -1.20946448 | $1.130 \times 10^{-10}$ | **PASS** | | |
| v | -0.71068205 | -0.71068205 | $5.254 \times 10^{-11}$ | **PASS** | | |
| c | -1.53788284 | -1.53788284 | $1.118 \times 10^{-10}$ | **PASS** | | |

### 6.4 Simultaneous Update & 100-Step Training Progression

All parameters are updated simultaneously using gradients evaluated exclusively at the original parameter values:

$$\theta_i^{(k+1)} = \theta_i^{(k)} - \eta \frac{\partial J}{\partial \theta_i} \Bigg|_{\{\theta^{(k)}\}}$$

Over 100 gradient descent steps with learning rate $\eta = 0.1$, the loss decreases monotonically from $J^{(0)} = 1.182542$ \to $J^{(10)} = 1.0211 \times 10^{-2}$ \to $J^{(100)} = 1.7026 \times 10^{-20}$, predicting $\hat{y} = 2.000000$ (matching target $y = 2.0$ to machine precision).

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## 7. Packaging, Reproducibility & AI Transparency

### 7.1 Standalone Windows 11 Release ( `CandidateID_Workbench.zip` )

- **Native Starter ( `Launch.exe` )**: Written in pure Win32 C++ (compiled with `cl.exe packaging/Launch.cpp /O2 /SUBSYSTEM:WINDOWS` ). Contains zero external DLL dependencies. - **Bundled Runtimes**: Qt 6.8.2 libraries deployed via `windeployqt` ; Microsoft Visual C++ CRT redistributables deployed alongside the application. - **Relocatable & Offline**: Tested by extracting to an isolated temporary directory and executing `Launch.exe` . The application opens immediately in offline standalone mode. - **Package Checksum**: Generated into `release/CandidateID_Workbench.sha256` .

### 7.2 AI Transparency & Ethical Engineering Compliance

In alignment with the assignment guidelines: - **AI Tool Utilization**: ChatGPT Codex / Google Antigravity served as an autonomous pair-programming assistant for boilerplate scaffolding, CMake script generation, and KaTeX formatting. - **Verification Integrity**: All mathematical formulas, physical equations, test matrices, and execution receipts were independently verified by the candidate engineer. Zero tests, benchmark values, or handwritten scans were fabricated. - **Audit Findings**: Discrepancies between previous AI text summaries and underlying source code were independently investigated, verified in C++ regression tests, and corrected.

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## 8. Conclusion

The **Unified Engineering Workbench** delivers a comprehensive, production-quality implementation of the PASS SCIENCE WORKS R&D Common Internship brief: - All 6 technical tasks (Tasks A–F) and mandatory gates are fully implemented, verified, and documented. - 100% of automated CTest test targets pass natively under MSVC on Windows 11 x64. - Clean, maintainable C++17 architecture with strict separation between numerical solvers and UI. - Self-contained, relocatable Windows 11 executable operating offline without external runtime dependencies. - All technical documentation, mathematical derivations, defect analyses, and verification receipts are published and preserved in the repository.